



PRINTED CIRCUIT BOARD

PART NUMBER

3530-0121-01

COMPONENT MAINTENANCE MANUAL WITH ILLUSTRATED PARTS LIST

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B/E Aerospace, Inc.
Commercial Aircraft Products Group
Beverage Maker Products
10800 Pflumm Road
Lenexa, KS 66215
TEL: 913-338-9800
FAX: 913-469-4032
CAGE: 63367

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25-33-32

Page T-1
21 Aug 2013
Rev 001

ALERT

THIS COMPONENT MAINTENANCE MANUAL IDENTIFIES REPLACEMENT PARTS AND REPAIR PROCEDURES SPECIFICALLY ENGINEERED FOR BEVERAGE MAKER PRODUCTS MANUFACTURED BY B/E AEROSPACE, INC., COMMERCIAL AIRCRAFT PRODUCTS GROUP. B/E PARTS ARE MANUFACTURED UNDER STRINGENT QUALITY CONTROL PROCESSES. USING NON-B/E PARTS MAY JEOPARDIZE YOUR WARRANTY AND RESULT IN DEGRADED PRODUCT PERFORMANCE. ALL MODIFICATIONS SHOULD BE COORDINATED IN ADVANCE TO AVOID THIS POSSIBILITY.

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COMPONENT MAINTENANCE MANUAL
3530-0121

REVISION NUMBER 001 DATED 21 AUGUST 2013

HIGHLIGHTS

This Component Maintenance Manual (CMM) with Illustrated Parts List (IPL) has been reviewed for technical changes resulting from B/E Service Bulletins, B/E Service Information letters, configuration changes and part improvements. Any existing Temporary Revisions (TRs) against this CMM have been incorporated with this revision.

This CMM and IPL have been completely revised and reformatted. Please replace the old copies of this manual with this Revision Number 001 dated 21 August 2013. A description of the changes follows.

SECTION AND PAGES	CHANGE DESCRIPTION
Title Page, T-1	Updated to reflect current revision.
Alert Page, A-1	Updated alert page to reflect current revision.
Service Bulletin Lists, SBL-1 and SBL-2	Updated Service Bulletin Lists to latest standards.
Service Information Letter Lists, SIL-1 and SIL-2	Updated List of Service Information Letter Lists to latest standards.
List of Effective Pages, LEP-1 and LEP-2	Updated List of Effective Pages to reflect current revision.
Introduction, INTRO-1 and INTRO-2	Updated Abbreviation Table.
Testing and Fault Isolation, 1001 thru 1006	Updated procedure in Functional test para 2.(A).(4) and added Diode (D11) in Table 1002.
Illustrated Parts List, 10001 thru 10010	Updated alpha/numeric index and deleted Vendor codes V18612, V1ZEC0, V33062, V60991, Items 25, 30, 40, 45 and 50 in illustration, and detailed parts list to reflect current revision.

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RECORD OF REVISIONS

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

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RECORD OF TEMPORARY REVISIONS

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

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SERVICE BULLETIN LIST

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

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SERVICE INFORMATION LETTER LIST

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COMPONENT MAINTENANCE MANUAL
3530-0121

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

LIST OF EFFECTIVE PAGES

<u>SECTION</u>	<u>PAGE</u>	<u>DATE</u>	<u>SECTION</u>	<u>PAGE</u>	<u>DATE</u>
Title	T-1	21 Aug 2013	Schematics and	2001	11 May 2012
Alert	A-1	21 Aug 2013	Wiring Diagrams	2002	11 May 2012
				2003	11 May 2012
				2004	11 May 2012
Highlights	HL-1	21 Aug 2013			
	HL-2	Blank	Disassembly	3001	11 May 2012
				3002	Blank
Record of	ROR-1	21 Aug 2013			
Revisions	ROR-2	Blank	Cleaning	4001	11 May 2012
				4002	Blank
Record of	RTR-1	11 May 2012			
Temporary	RTR-2	Blank	Inspection/Check	5001	11 May 2012
Revisions				5002	Blank
Service Bulletin	SBL-1	21 Aug 2013	Repair	6001	11 May 2012
List	SBL-2	Blank		6002	Blank
			Assembly	7001	11 May 2012
Service Information	SIL-1	21 Aug 2013		7002	Blank
Letter List	SIL-2	Blank			
			Special Tools,	9001	11 May 2012
List of Effective	LEP-1	21 Aug 2013	Fixtures, and	9002	11 May 2012
Pages	LEP-2	Blank	Equipment	9003	11 May 2012
				9004	Blank
Table of Contents	TOC-1	11 May 2012			
	TOC-2	Blank	Storage	15001	11 May 2012
				15002	Blank
Introduction	INTRO-1	21 Aug 2013			
	INTRO-2	11 May 2012	Illustrated Parts	10001	21 Aug 2013
			List	10002	21 Aug 2013
Description and	1	11 May 2012		10003	21 Aug 2013
Operation	2	11 May 2012		10004	21 Aug 2013
				10005	21 Aug 2013
				10006	21 Aug 2013
Testing and Fault	1001	11 May 2012		10007	21 Aug 2013
Isolation	1002	21 Aug 2013		10008	11 May 2012
	1003	21 Aug 2013		10009	Blank
	1004	21 Aug 2013		10010	21 Aug 2013
	1005	21 Aug 2013			Blank
	1006	Blank			

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
INTRODUCTION	INTRO-1
DESCRIPTION AND OPERATION	1
TESTING AND FAULT ISOLATION	1001
SCHEMATIC AND WIRING DIAGRAMS	2001
DISASSEMBLY	3001
CLEANING	4001
INSPECTION/CHECK	5001
REPAIR	6001
ASSEMBLY	7001
FITS AND CLEARANCES	8001 (Not Applicable)
SPECIAL TOOLS, FIXTURES, AND EQUIPMENT	9001
SPECIAL PROCEDURES	11001 (Not Applicable)
REMOVAL	12001 (Not Applicable)
INSTALLATION	13001 (Not Applicable)
SERVICING	14001 (Not Applicable)
STORAGE	15001
REWORK	16001 (Not Applicable)
ILLUSTRATED PARTS LIST	10001

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

INTRODUCTION

1. General

- A. This manual has been prepared to fulfill the requirements of ATA iSpec 2200 for Component Maintenance Manuals.
- B. All procedures established in this manual have been performed and verified acceptable by trained personnel at B/E Aerospace, Inc. A competent shop mechanic who is unfamiliar with these units should, with a minimum amount of effort and tools, be able to isolate the problems of a nonfunctional unit and restore it to service.
- C. Procedures are provided for the full range of maintenance and include all the work functions that could or may be done to the unit throughout its entire service life.
- D. A competent shop mechanic should use this manual first to test and isolate a faulty component. The mechanic should then go directly to the applicable manual subject to effect a repair in the specific problem area without disturbing other areas.

2. Abbreviations

The following abbreviations may be used in this manual:

ABBREVIATION	MEANING
ASSY	Assembly
ATA	Air Transport Association
° C	Degree Celsius or Centigrade
CAGE	Commercial and Government Entity
EFF	Effectivity
° F	Degree Fahrenheit
G	Grams
Hz	Hertz
IPL	Illustrated Parts List
LB	Pound
MAX	Maximum
MF	Micro Farad

ABBREVIATION	MEANING
MM	Millimeter
NO	Number
PCB	Printed Circuit Board
PN	Part Number
REQ	Required
RF	Reference
SB	Service Bulletin
SIL	Service Information Letter
SUPSD	Superseded
SUPSDS	Supersedes
TTL	Total
VAC	Voltage, Alternating Current

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

3. Verification

The sections of this publication shown below have been verified on the date(s) specified. Where actual performance of the functions (tasks) cannot be done prior to issuing a manual, the function (tasks) must be verified by simulation. Any function (task) verified by simulation must be verified by actual performance at the earliest possible time.

VERIFIED SECTION	VERIFIED BY	VERIFIED DATE
Testing and Fault Isolation	Simulation	11-Jan-2012
Disassembly	Simulation	11-Jan-2012
Assembly	Simulation	11-Jan-2012

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

DESCRIPTION AND OPERATION

1. General

- A. The Printed Circuit Board, PN 3530-0121-01 is supplied by B/E Aerospace, Inc., Commercial Aircraft Products Group, Lenexa, Kansas.

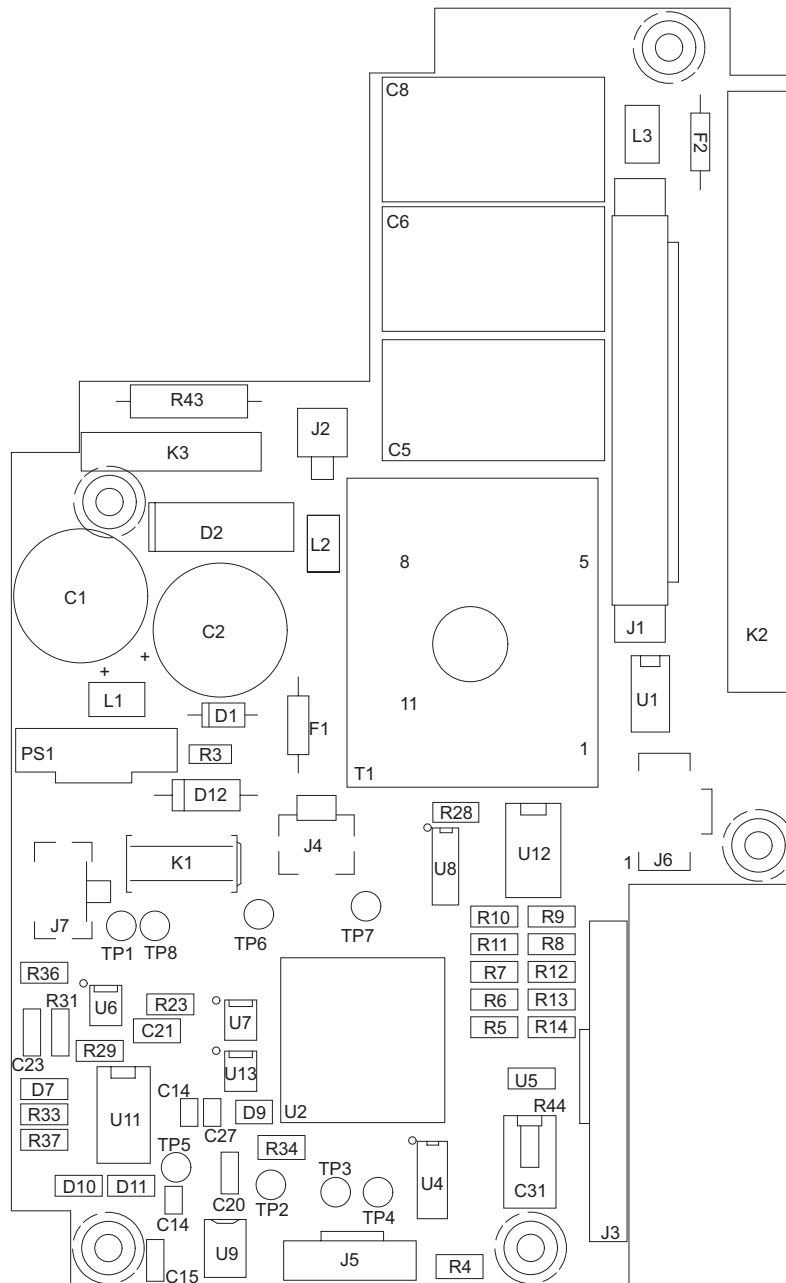


Figure 1
Printed Circuit Board

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

2. Description

- A. The Printed Circuit Board assembly is software controlled microcontroller hardware for the purpose of controlling the Endura 4510 and 4511 beverage makers.
- B. The Printed Circuit Board Features a solid state heater relay and opto-isolated phase loss detection in the event of overtemperature situation.
- C. The Printed Circuit Board is equipped with server detection, server level detection and a low water tank detection circuit.

3. Leading Particulars

- A. The Printed Circuit Board meets the following performance specifications.

Table 1
Leading Particulars

ITEM	CHARACTERISTIC
Electrical Input Range	95.0 to 130.0 VAC, 400 Hz
Wattage	30 Watts
Dimensions	6.8 inch x 4.2 inch x 0.5 inch (172.7 mm x 106.7 mm x 12.7 mm)
Weight	0.66 lb (300 g) Max
Operating Temperature Range	35° F to 131° F (1.67° C to 55° C)

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

TESTING AND FAULT ISOLATION

1. General

- A. TESTING AND FAULT ISOLATION section covers the test equipment, bench setup, functional checkout procedure and fault isolation procedures required to return an inoperative Printed Circuit Board to service.
- B. Completely read and thoroughly understand the entire TESTING AND FAULT ISOLATION section. Become thoroughly familiar with the test sequences and their interactive correlations before performing any tests.
- C. The test equipment and materials required for testing of the Printed Circuit Board are listed in Table 1001.

NOTE: Equivalent substitutes may be used for listed items.

Table 1001
Test Equipment and Materials

NOMENCLATURE	PARTS/SPEC NO.	SOURCE (CAGE)
K2 Relay Allignment Fixture (refer to Figure 9001)	997058	B/E Aerospace, Inc (V63367)
Load Box	997314 (7.5 amp) or 997937 (1 amp)	B/E Aerospace, Inc (V63367)
P.C.Board Test Fixture	997113	B/E Aerospace, Inc (V63367)
Power Supply 115 VAC, 3 Phase, 400 Hz	None	Commercially Available

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

2. Procedure

A. Functional Test

- (1) Make sure that the P.C. Board Test Fixture (refer to Table 1001) is in the following condition and connect the load box (refer to Table 1001) according to the wire labels on the load box (refer to Table 1001) and connections on the test fixture. The load box (refer to Table 1001) is used to make sure that the K2 Solid State Relay is operating correctly.

SWITCH	CONDITION
Main Power	OFF or Open
"C" Phase Heater	Closed
Tank	Full
Server	Empty
Optical	Empty
Brew Handle	Down
Temperature Simulator	84° F

- (2) Connect the P.C. Board Test Fixture and the load box (refer to Table 1001) to a 115 VAC 3 phase 400 Hz power supply.
- (3) Place the P.C. Board under test onto the stand offs of the test fixture and connect the six cables to the associated connectors on the P.C. Board (Example: Cable P1 to Connector J1).
- (4) Move the "Main Power" switch to "ON" and make sure that NO other action takes place on the test fixture. Make sure that P.C Board is communicating with the micro-controller by rotating the "Temperature Simulator" and observing change in "Temperature" window of the program. Reset "Temperature Simulator" to 84° F.
- (5) Press and release the "Power" switch and make sure that the "Power" indicator along with the three "Heater" indicators illuminate. If the "Power" switch indicator fails to illuminate, see "No Power" in the Fault Isolation Table. If the "Power" switch indicator is in a triple flash mode, see "Continuous Triple Flash mode" in the Fault Isolation Table. If any of the "Heater" indicators fail to illuminate, see "Missing Heaters" in the Fault Isolation Table. If the "Platen Heater" indicator illuminates, see "Platen Heater Continuous" in the Fault Isolation Table. If the "Low Water" Indicator illuminates, see "Continuous low water mode" in the Fault Isolation Table. If any other indicator is illuminated, the board is faulty and not field serviceable.
- (6) Rotate the "Temperature Simulator" counter clockwise towards "42° F" mark and make sure that the three "Heater" indicators stay illuminated until approximately the "42° F" mark and fully OFF at the "32° F" mark. If any of the "Heater" indicators extinguish prematurely or do not extinguish, the board is faulty and not field serviceable.
- (7) Rotate the "Temperature Simulator" clockwise to the "84° F" mark and hold. The three "Heater" indicators should be illuminated.

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

- (8) Move the "C Phase Heater" switch to "Open" and the "C" Heater Indicator will extinguish. Make sure that in approximately 3 seconds, the "A" and "B" Heater Indicators extinguish and the "Power" indicator begins to triple flash. If the "A" and "B" heater indicators do not extinguish, see "No Triple Flash" in Fault Isolation Table.
- (9) Move the "C Phase Heater" switch to "Close" and press the "Power" button to OFF and again to ON. Make sure that all the three "Heater" Indicators are illuminated.
- (10) Rotate the "Temperature Simulator" clockwise towards the "190° F" mark and make sure that the three "Heater" indicators stay illuminated until approximately the "190° F" mark where they will pulsate until approximately the "196° F" mark where they should extinguish completely. If the indications are incorrect, the board is faulty and not field serviceable.
- (11) Continue to rotate the "Temperature Simulator" clockwise until approximately the "235° F" mark and make sure that the over temperature protection is activated evident by the Double Flash of the "Power" switch indicator. The three "Heater" indicators will extinguish as well. If there is no Double Flash, the board is faulty and not field serviceable.
- (12) Rotate the "Temperature Simulator" counter clockwise until approximately "84° F". Press the "Over Temperature Reset" switch located on the solder side of the PC Board and then the "Power" switch OFF and ON again. Make sure that the three "Heater" indicators illuminate and that the "Power" switch indicator remains in a constant ON state. If the unit doesn't return to normal operation, see Safety Latch Reset Switch Failure in the Fault Isolation Table.
- (13) Press and release the "Brew" switch and make sure that the "Brew" indicator illuminates and stays illuminated along with NO other action taking place. If the "Brew" switch Indicator does not illuminate, the board is faulty and not field serviceable.
- (14) Press the "Tea" switch and make sure that no other action takes place.
- (15) Press the "Platen" switch and make sure that both the "Platen" switch indicator and "Platen Heater" indicator illuminate. If the "Platen" switch indicator does not illuminate, the board is faulty and not field serviceable. If the Platen Heater indicator does not illuminate, see No Platen Heater in the Fault Isolation Table. Move the "Platen" switch to the OFF position.
- (16) Rotate the "Temperature Simulator" clockwise until the "Brew Solenoid" indicator activates. Make sure that the "IRED" indicator blinks ON and OFF. If the "Brew Solenoid" indicator does not illuminate, the board is faulty and not field serviceable.
- (17) The "Brew Solenoid" indicator will extinguish after approximately 10 seconds and will remain extinguished for approximately 10 seconds. The next three steps can be accomplished after the "Brew Solenoid" illuminates a second time.
- (18) Move the "Brew Handle" switch to the "UP" position and make sure that the "Brew Solenoid" indicator extinguishes. Move the "Brew Handle" switch to the "Down" position and make sure that the "Brew Solenoid" indicator illuminates. After approximately 10 seconds, the aroma control feature will deactivate and the "IRED" indicator illuminates at 100% duty. If the "Brew Solenoid" indicator or "IRED" indicator does not respond correctly, the board is faulty and not field serviceable.

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

- (19) Press the "Hot Water" switch and make sure that the "Hot Solenoid" indicator illuminates and that the "Brew Solenoid" indicator extinguishes. Release the switch and the "Hot Solenoid" indicator should extinguish and the "Brew Solenoid" indicator should illuminate. If the "Hot Solenoid" indicator does not illuminate, the board is faulty and not field serviceable.
- (20) Press the "Cold Water" switch and make sure that the "Cold Solenoid" indicator illuminates and that the "Brew Solenoid" indicator extinguishes. Release the switch and the "Cold Solenoid" indicator should extinguish and the "Brew Solenoid" indicator should illuminate. If the Cold Solenoid indicator does not illuminate, the board is faulty and not field serviceable.
- (21) Move the "Optical" switch to the "Full" position and make sure that the "Brew Solenoid" and that the "IRED" indicator extinguishes. Also, make sure that make sure 10 seconds after the "Brew Solenoid" indicator extinguishes, the "Brew" indicator extinguishes. If the "Brew Solenoid" indicator or "Brew" switch indicator do not respond appropriately, the board is faulty and not field serviceable.
- (22) Move the "Brew Handle" switch to the "UP" position and back to the "Down" position, rotate the "Temperature Simulator" to the "84° F" mark, and move the "Optical" switch to the "Empty" position.
- (23) Move the "Tank" switch to the "Empty" position and make sure that the three "Heater" indicators deactivate along with the "Brew Solenoid" and "Low Water" indicators illuminate. Move the "Tank" switch to the "Full" position and make sure that the three "Heater" indicators reactivate along with the "Brew Solenoid" and "Low Water" indicators deactivating. If the "Heater" indicators do not extinguish or the "Low Water" indicator does not illuminate, see "No Low Water Protection" in the Fault Isolation Table.
- (24) Press and release the "Tea" switch. Make sure that the "Tea" indicator illuminates and NO other action takes place. If the "Tea" indicator does not illuminate, the board is faulty and not field serviceable.
- (25) Move the "Temperature Simulator" clockwise until the "Tea Solenoid" indicator illuminates. If the "Tea Solenoid" indicator does not illuminate, the board is faulty and not field serviceable.
- (26) Move the "Brew Handle" switch to the "UP" position and make sure that the "Tea Solenoid" indicator extinguishes. Move the "Brew Handle" switch to the "Down" position and make sure that the "Tea Solenoid" indicator illuminates. If the "Tea Solenoid" or "Tea" switch lights do not respond appropriately, the board is faulty and not field serviceable.
- (27) Move the "Server" switch to "Full" and make sure that the "Tea Solenoid" and "Tea" switch indicators extinguish. Move the "Server" switch to the "Empty" position. If the indicators do not deactivate, see Backup Probe Failure in the Fault Isolation Table.
- (28) Press and release the "Power" switch and make sure that the three "Heater" indicators deactivate and the "Power" indicator extinguishes.
- (29) Testing is complete. Move the "Main Power" switch to "OFF" and ensure NO other action takes place. Remove the board from the test fixture.
- (30) If all of the steps above are in compliance, stamp the board with an "FT". This signifies that the board has passed the functional test. The stamp should be with black or white permanent ink.

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

3. Fault Isolation

- A. Faults most likely encountered during testing or operation of the Printed Circuit Board are listed in Table 1002, together with possible causes, isolation procedures, and corrective action. Component reference designators will be used to help follow the System Wiring Diagram, refer to Figure 2001 of SCHEMATICS AND WIRING DIAGRAMS section.

NOTE: The lamp that illuminates the POWER button is also used to indicate certain faults by flashing OFF and ON as described in Table 1002.

Table 1002
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTION
No Power	F1 Open and D12 Shorted	Replace with new.
	F1 Open and D2 Shorted	Replace with new.
Low Water Mode Continuous	D8 Shorted	Replace with new.
No Low Water Protection	D6 and D7 Failure	Replace with new.
Missing Heaters	K2 Relay Inoperative	Replace with new.
	K1 Relay Defective	Replace with new.
Safety Latch Reset Switch Failure	SW1 Defective	Replace with new.
Backup Probe Failure	D10, D11, and D12 Defective	Replace with new.
Triple Flash Mode Continuous	U1 and/or K2 Defective	Replace with new.
No Triple Flash Mode	U1 Defective	Replace with new.
No Platen Heater	K3 Inoperative	Replace with new.
Platen Heater Continuous	K3 Shorted	Replace with new.

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

SCHEMATICS AND WIRING DIAGRAMS

1. General

- A. This section contains the wiring diagram for the Printed Circuit Board.
- B. Refer to Figure 2001 for Schematics of Printed Circuit Board.

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

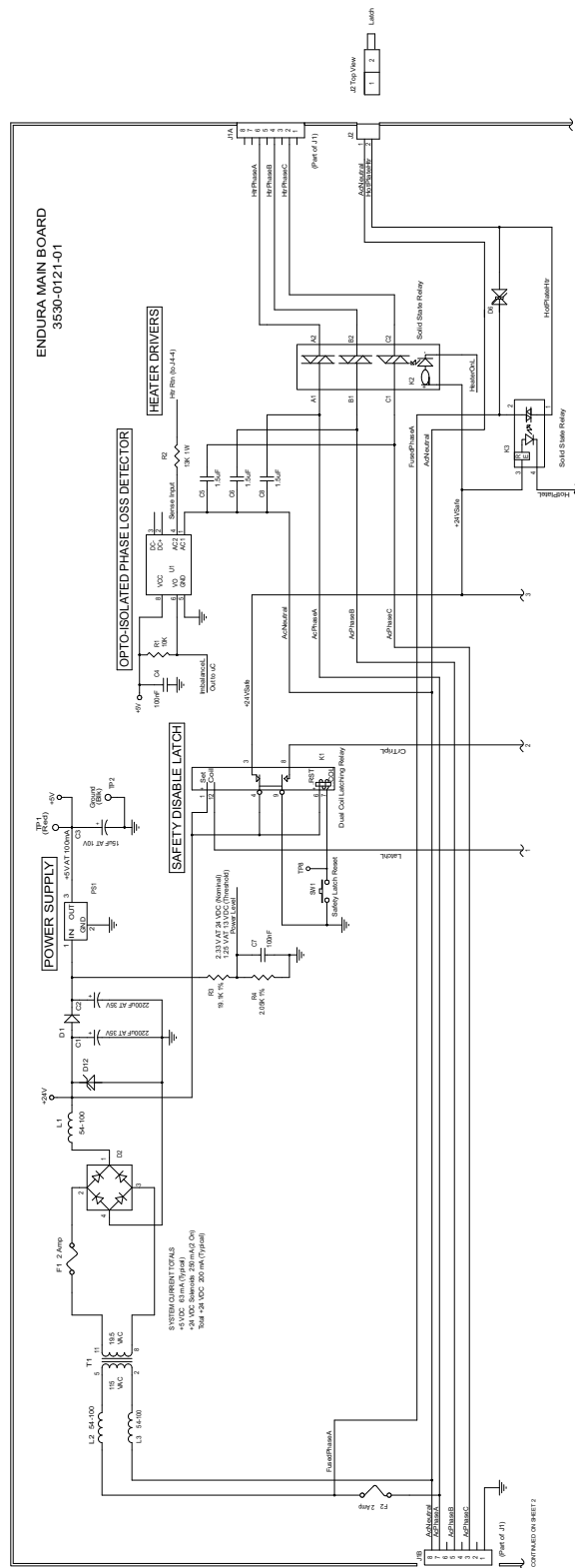


Figure 2001 (Sheet 1 of 3)
Wiring Schematic

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25-33-32

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

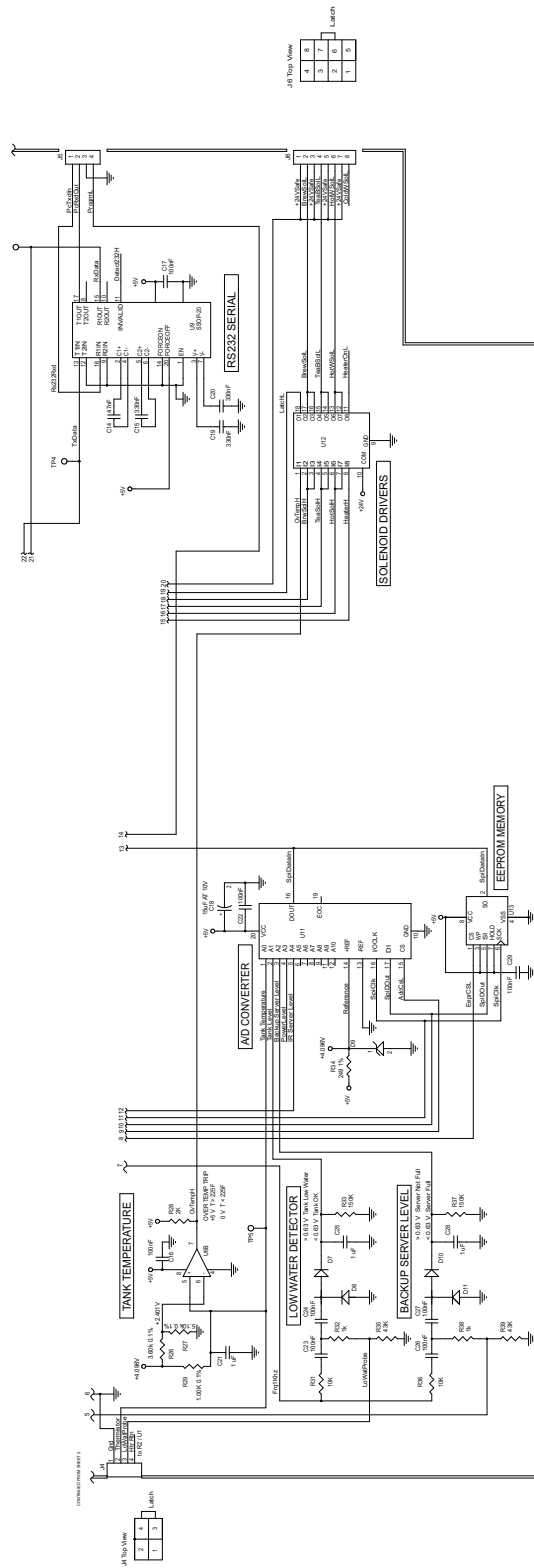


Figure 2001 (Sheet 3 of 3)
Wiring Schematic

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COMPONENT MAINTENANCE MANUAL
3530-0121

DISASSEMBLY

1. General

WARNING: THE PRINTED CIRCUIT BOARD IS A STATIC SENSITIVE DEVICE. ELECTROSTATIC PROTECTION IS REQUIRED.

2. Procedure

- A. The components of the Printed Circuit Board must be desoldered for disassembly (Per PCS-1001 and/or IPC-J-STD-001D and/or IPC-A-610 Class 2).

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

CLEANING

1. General

- A. The circuit card shall be free of dust, debris and other contamination that could interfere with the function of the electronic components.

2. Procedure

- A. Clean the printed circuit board by blow drying with filtered, oil-free dry air.

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COMPONENT MAINTENANCE MANUAL
3530-0121

INSPECTION/CHECK

1. Procedure

A. Visual Inspection

- (1) Inspect the printed circuit board to make sure it is free of dust, fingerprints, or other visible contamination.
- (2) Examine the printed circuit board for scratches or other visible damage.
- (3) Examine the printed circuit board to make sure that the correct identification is on the assembly.

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

REPAIR

1. General

- A. Repair of the Printed Circuit Board is limited to replacement of broken or damaged parts as listed in the ILLUSTRATED PARTS LIST Section.

2. Procedure

NOTE: To remove existing Humiseal in order to make repairs. (Use Humiseal Stripper 1080.)

- A. Repair Printed Circuit Board per soldering specification PCS-1001 and/or IPC J-STD-001D and/or IPC-A-610 Class 2 except as noted for K2 Relay.
- B. After completion of solder process for installed electronic components clean PC board to remove flux residue with de-natured alcohol.
- C. Apply conformal coat as noted:
 - (1) Apply a uniform coating .001 to .006 thick of Humiseal 1B31 conformal coating per IPC-CC-830 using a disposable brush.
 - (2) Connector pins, Switches, I.C. Socket Test Points, Fuse Eyelets and K2 Relay Heat Sink shall be free of Humiseal Coating.
 - (3) Let Humiseal dry per manufacturer's recommendations.

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

ASSEMBLY

1. General

WARNING: THE PRINTED CIRCUIT BOARD IS A STATIC SENSITIVE DEVICE. ELECTROSTATIC PROTECTION IS REQUIRED.

2. Procedure

- A. Refer to repair section soldering specification for assembly of Printed Circuit Board (Per PCS-1001 and/or IPC-J-STD-001D and/or IPC-A-610 Class 2).

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

SPECIAL TOOLS, FIXTURES, AND EQUIPMENT

1. General

- A. All work described in this manual may be conducted in a normal work area.
- B. The special tools, fixtures, and equipment required for maintenance of the Printed Circuit Board are listed in Table 9001.

NOTE: Equivalent substitutes may be used for listed items.

Table 9001
Special Tools, Fixtures, and Equipment

NOMENCLATURE	PARTS/SPEC NO	SOURCE (CAGE)
K2 Relay Alignment Fixture	997058	B/E Aerospace, Inc (V63367)
Load Box	997314 (7.5 amp) or 997937 (1 amp)	B/E Aerospace, Inc (V63367)
P.C.Board Test Fixture	997113	B/E Aerospace, Inc (V63367)
Power Supply 115 VAC, 3 Phase, 400 Hz	None	Commercially Available

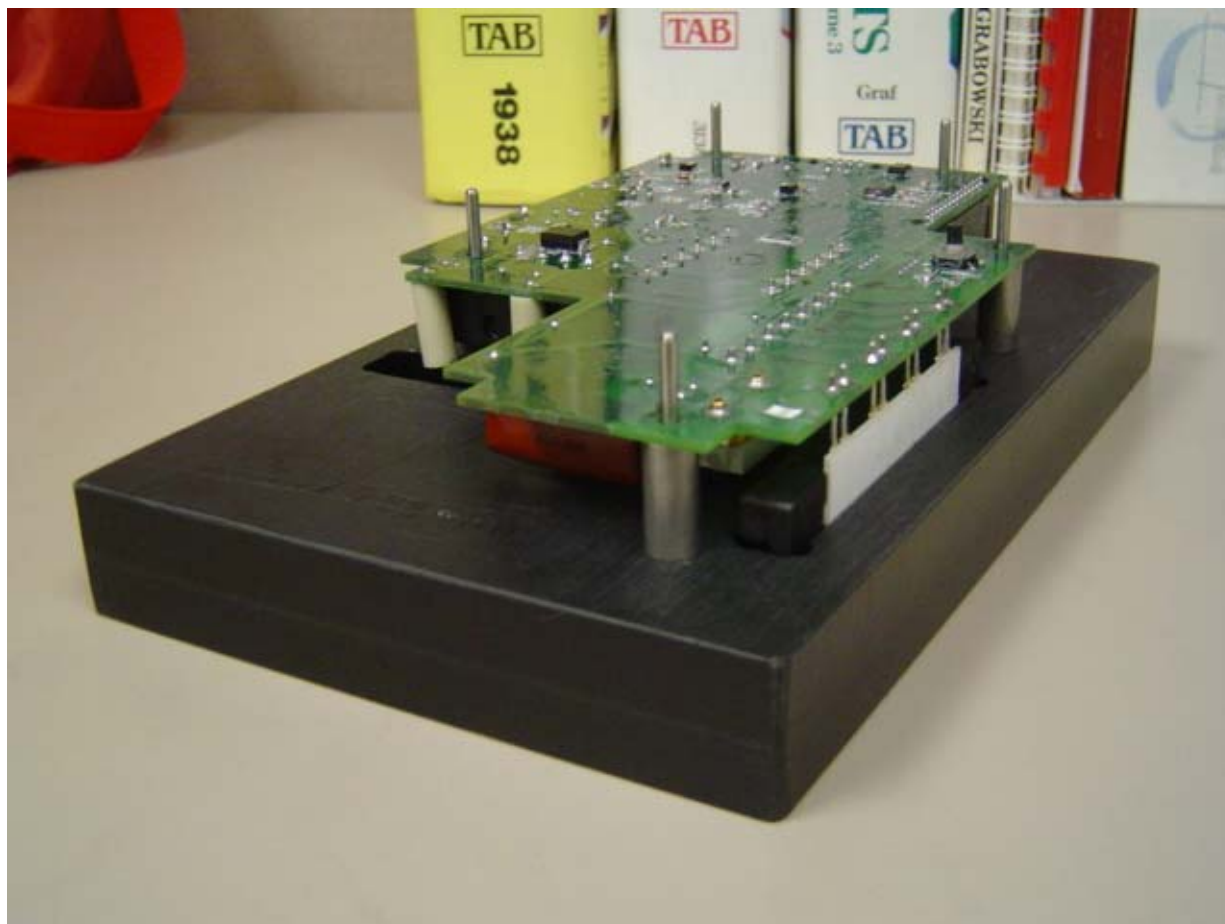


Figure 9001
K2 Relay Alignment Fixture (Sheet 1 of 2)



Figure 9001
K2 Relay Alignment Fixture (Sheet 2 of 2)

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

STORAGE

1. General

A. The Printed Circuit Board shall be properly stored when not in use.

2. Procedure

A. Place the Printed Circuit Board in an anti-static bag that meets or exceeds MIL-PRF-81705.

B. For prolonged storage, the entire assembly should be protected by sealing it in a polyethylene plastic bag or by placing it in a clean, cardboard container.

C. Mark the packaging with component nomenclature and part number.

D. Store in a clean, dry location.

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

ILLUSTRATED PARTS LIST

1. Introduction

The ILLUSTRATED PARTS LIST is intended for use in provisioning, storing, and issuing replaceable parts for the component, and in identification of new and reclaimed parts.

2. Vendor's Parts

Vendor's parts used in the Printed Circuit Board and which are not altered by B/E Aerospace, Inc. are listed in the Detailed Parts List by the vendor's part number, vendor's description of the part, and vendor's code listed in parentheses following the description. Vendors' code symbols used in this publication are taken from the CAGE Cataloging Handbook, and consist of the applicable code symbol preceded by the Letter "V". Table 10001 shows a numerically arranged list of vendor codes used in this publication.

Table 10001
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V09353	C and K Components Inc. 57 Stanley Avenue Watertown, MA 02472
V0BED2	Fairchild Semiconductor 1801 Mc Carthy Blvd Milpistas, CA 95035
V0J4G8	Panasonic Industrial Company Affiliation of Matsushita Electric Corporation Of America 2 Panasonic Way Secaucus, NJ 07094
V11532	Teledyne Technologies Inc. DBA Teledyne 12525 Daphne Hawthorne, CA 90250
V12060	Diodes Inc. 3050 East Hillcrest Drive Westlake Village, CA 91632
V1NGH3	Fairchild Semiconductor International Inc. Division Discrete Power Products 65 East Nasa Boulevard Suite 101 Melbourne, FL 32901
V6CDP2	Omron Electronics Components LLC OCB-AM 55 East Commerce Drive Suite B Schaumburg, IL 60173
V63367	B/E Aerospace, Inc. 10800 Pflumm Road Lenexa, KS 66215

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V75915	Littlefuse Inc. 8755 West Higgins Road Suite 500 Chicago, IL 60631
V7K104	World Products Inc. 19654 8th Street East Sonoma, CA 95476

3. Use of the Illustrated Parts List

A. When the part number is known:

Turn to the Numerical Index and locate the part number. Figure and item number on the illustration where the part appears are listed in the column to the right of the part number. Turn to the illustration and locate the item number. The corresponding item number in the accompanying Detailed Parts List will give part number, description, assembly relationship, quantity required for that particular application, and whether or not the part appears in the illustrations.

B. When the part number is not known:

Look through the figures and identify the part by appearance. Note the figure number and the item number. The corresponding Fig-Item number in the Detailed Parts List shows the part number, description, assembly relationship, effectivity code, and the number of units per assembly.

4. Material Arrangement

A. Numerical Index

The Numerical Index is an alphanumeric listing of all the part numbers in the Detailed Parts List. Each part number is listed together with its corresponding item number, ILLUSTRATED PARTS LIST figure number, and the total required number of units per assembly. The figure-item number is composed of two parts. To the left of the dash is the figure number. It is the number of the figure in the ILLUSTRATED PARTS LIST that shows the part. To the right of the dash is the item number. It is the number that identifies a part in a figure.

B. Illustrations

The ILLUSTRATED PARTS LIST illustrations interrelate to the Numerical Index and the Detailed Parts List. The separate illustrations pictorially describe recommended disassembly/assembly sequence and support DISASSEMBLY and ASSEMBLY as well as the Numerical Index and the Detailed Parts List of this manual.

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

C. Detailed Parts List

(1) Fig-Item

The Detailed Parts List is arranged numerically by item numbers. These are the same item numbers listed in the Numerical Index. In the Detailed Parts List the figure number portion of the figure item number is listed only once per column. Immediately below a figure number are the item numbers that appear on that figure. A dash in front of an item number means the corresponding part is not shown in that or any other figure. Each item number is listed together with its corresponding part number, nomenclature (description, assembly relationship), effectivity code, and units per assembly.

(2) Part Number

The part number column contains the original manufacturer's part number. When standard parts are used, the standard part number is listed in this column. When B/E Aerospace, Inc. is not the original manufacturer the vendor's part number is shown in the part number column, and the CAGE vendor number and equivalent B/E Aerospace, Inc. part number are given in the nomenclature column.

(3) Airline Part Number

This column is left blank for airline internal use.

(4) Nomenclature

(a) Indenture System

The indenture system shows the relationship of parts and assemblies to next higher assemblies or installations, as follows:

1 2 3 4 5 6 7

Unit (Product which this manual describes)

- Detail parts for assembly into the unit

- Subassemblies

- Attaching parts for subassemblies

* * *

- • Detail parts for subassemblies

- • Sub-subassemblies

- • Attaching parts for sub-subassemblies

* * *

- • • Detailed parts for sub-subassemblies

(b) Attaching Parts

Attaching parts are captioned ATTACHING PARTS and are listed immediately following the parts attached. The * * * symbol follows the last item of the attaching parts group.

(c) Vendor Code

Parts manufactured by companies other than B/E Aerospace, Inc. are identified by an appropriate vendor's CAGE code following the nomenclature, and are preceded by the letter "V". Standard parts such as AN, MS, etc., are not identified by a vendor code.

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

(5) Effectivity Code

When two or more units (main assemblies) are listed in the same ILLUSTRATED PARTS LIST, a code letter ("A", "B", "C", etc.) is assigned to each main unit. All subcomponents that are peculiar to a particular unit are identified by the same effectivity code letter as the unit. If parts are common to all units, the effectivity code column is left blank.

(6) Units per Assembly

Quantities specified in the "UNITS PER ASSY" column are the total number of each part required per assembly or subassembly and are not necessarily the total used per the complete unit.

5. Parts Replacement Data

Interchangeability relationship between parts is identified in the "NOMENCLATURE" column. A list of terms used and their definition is as follows.

TERM	ABBREVIATION	DEFINITION
Alternate	ALT	Part fully meets functional and structural specifications, but differs either in overall external dimensions, connection installation, and/or mounting provisions, and may require additional rework or modifications to install in a specific application.
Optional	OPT	The part is fully interchangeable in form, fit, and function with other item numbers shown.
Preferred	PRFD	Part is preferred over the other optional parts shown.
Replaced By	REPLD BY	Part is replaced by and is interchangeable with item number listed in notation.
Replaces	REPLS	Part replaces and is interchangeable with number listed in notation.
Superseded By	SUPSD BY	Part is replaced by and is not interchangeable with item number listed in notation.
Supersedes	SUPSDS	Part replaces and is not interchangeable with item number listed in notation.
Service Bulletins	PRE SB POST SB RWK SB	Identifies when a service bulletin (SB) has an effect on a part. PRE shows the condition of the part before the work in the service bulletin is done; POST shows the condition after the work is done; RWK shows that the part is reworked by the service bulletin.

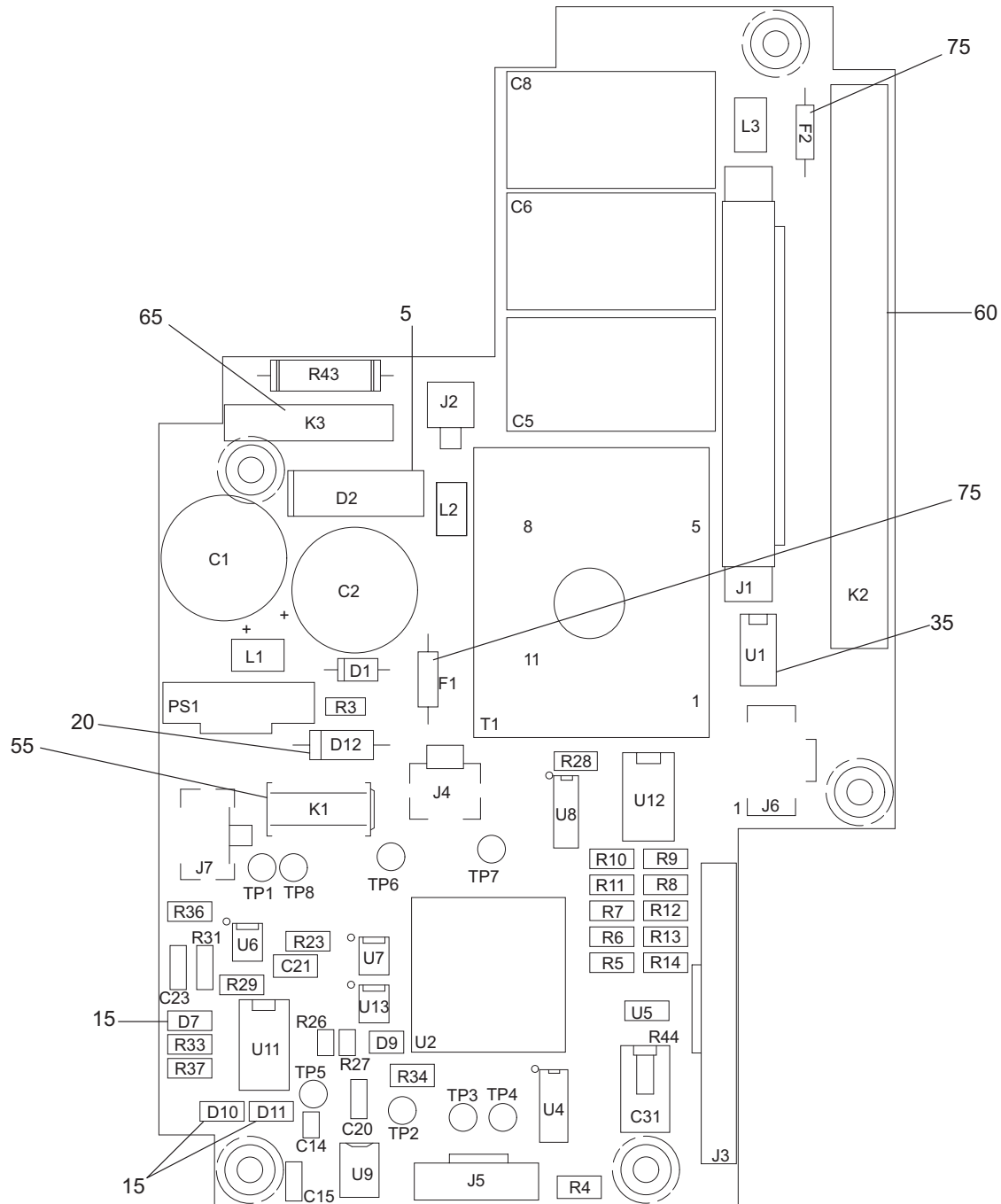
B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

6. Alpha/Numerical Index

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
B32924C3155K		1	-25	DEL
C3P24D25		1	60	1
EE2-24TNUX		1	55	1
ERZVF2M241		1	10	1
FDLL4148		1	15	4
G3MC201PDC24		1	65	1
HPCL3700		1	35	1
KSC641G		1	70	1
P6KE33AT		1	20	1
RS404L		1	5	1
RSO2B220R0		1	-30	DEL
25C160E/SN		1	-45	DEL
3530-0121-01		1	-1	RF
3530-0270-01		1	-40	DEL
473002		1	75	2
54-100-3		1	-50	DEL

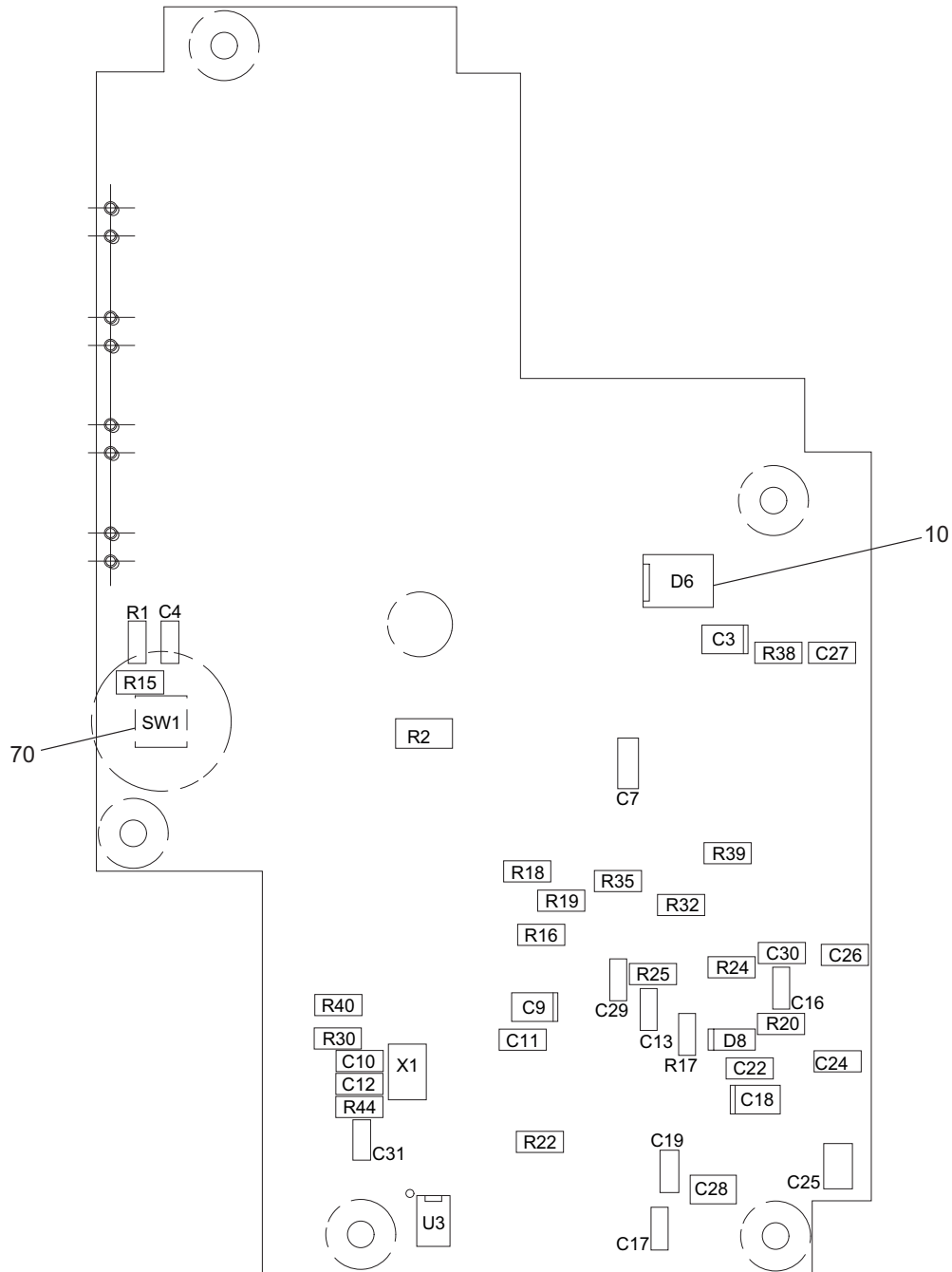
- ITEM NOT ILLUSTRATED

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121



IPL Figure 1 (Sheet 1 of 2)
Printed Circuit Board

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121



IPL Figure 1 (Sheet 2 of 2)
Printed Circuit Board

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B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-1	3530-0121-01		ASSEMBLY, PRINTED CIRCUIT BOARD		RF
5	RS404L		• DIODE BRIDGE (D2) (V12060) (B/E AEROSPACE STOCK CODE RC-50154)		1
10	ERZVF2M241		• DIODE, TRANSORB (D6) (V0J4G8) (B/E AEROSPACE STOCK CODE RC-50155)(TPN ERZ-VF2M241)		1
15	FDLL4148		• DIODE, SIGNAL (D7, D8, D10, D11) (V0BED2) (B/E AEROSPACE STOCK CODE RC-50138)		4
20	P6KE33AT		• DIODE, TRANSORB, 33 V (D12) (V12060) (B/E AEROSPACE STOCK CODE RC-50153) (TPN P6KE33A-T)		1
-25	B32924C3155K		DELETED		DEL
-30	RSO2B220R0		DELETED		DEL
35	HPCL3700		• IC OPTOCOUPLER (U1) (V1NGH3) (B/E AEROSPACE STOCK CODE IC-50061) (TPN HPCL-3700)		1
-40	3530-0270-01		DELETED		DEL
-45	25C160E/SN		DELETED		DEL
-50	54-100-3		DELETED		DEL
55	EE2-24TNUX		• RELAY, MAGNETIC LATCHING (K1) (V7K104) (B/E AEROSPACE STOCK CODE RL-50066)		1
60	C3P24D25		• RELAY, SOLID STATE, 3 PHASE (K2) (V11532) (B/E AEROSPACE STOCK CODE RL-50063)		1
65	G3MC201PDC24		• RELAY, SOLID STATE, 1 PHASE (K3) (V6CDP2) (B/E AEROSPACE STOCK CODE RL-50068) (TPN G3MC-201P-DC24)		1
70	KSC641G		• SWITCH, TACT (SW1) (V09353) (B/E AEROSPACE STOCK CODE SW-50292)		1
75	473002		• FUSE, 2 AMP (F1, F2) (V75915) (B/E AEROSPACE STOCK CODE FU-50208)		2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE
COMPONENT MAINTENANCE MANUAL
3530-0121

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